EPFL - Computer Science

CS-401 Applied Data Analysis

Lecture 04 — October 1, 2025

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Describing Data

Overview of today's lecture

- Part 1: Descriptive statistics
- Part 2: Quantifying uncertainty
- Part 3: Relating two variables

Part 1: Descriptive statistics

Robust statistics

A statistic is said to be robust if it is not sensitive to **extreme values**

Min, max, mean, standard deviation are not robuts

Median, quartiles (and others) are robust

Heavy-tailed distributions

- Some distributions are all about the extreme values
- E.g., power laws: $f(x) = ax^{-k}$
  - Very very large values are, "but not very rare"
  - Body size vs. city size
  - For k ¡= 3: infinite variance
  - For k ¡= 2: infinite variance, infinite mean
  - Don't report (arithmetic) mean/variance for power-law-distributed data!
  - Use robust statistics (e.g., mean, quantiles, etc.) or geometric mean

Generalized means

- Common trick: transform data into a different space (via function f), take mean there, then transform back into the original space (via f^{-1})

$$f^{-1}(\frac{1}{n}\sum_{i=1}^{n} f(x_i)).$$

- f(x) = x^2, $f^{-1}(x) = \sqrt{x}$ "root mean square"
- f(x) = x, $f^{-1} = x$ "arithmetic mean"
- f(x) = log(x), f^{-1} = exp(x) "geometric mean"
- f(x) = 1/x, $f^{-1} = 1/x$ "harmonic mean"

Distributions

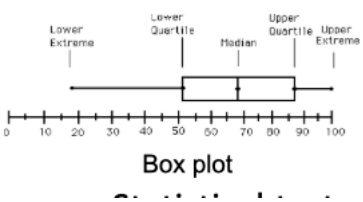
Some important distributions:

- **Normal**: see previous slides
- **Poisson**: the distribution of counts that occur at a certain "rate"; e.g., number of visits to a given website in a fixed time interval.
- **Exponential**: the interval between two such events
- **Binomial/multinomial**: the number of "successes" (e.g., coin flips = heads) out of n trials
- **Power-law/Zipf/Pareto/Yule**: e.g., frequencies of different terms in a document; city size

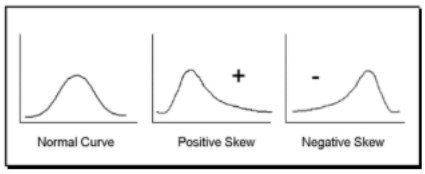
You should understand the distribution of your data before applying any model!

"Dear data, where are you from?"

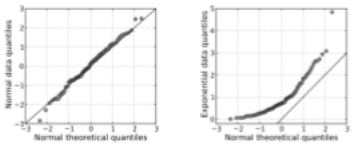
- Visual inspection for ruling out certain distributions: e.g., when histogram/box plot is asymmetric (even for large sample size), the data cannot be normal



Box plot



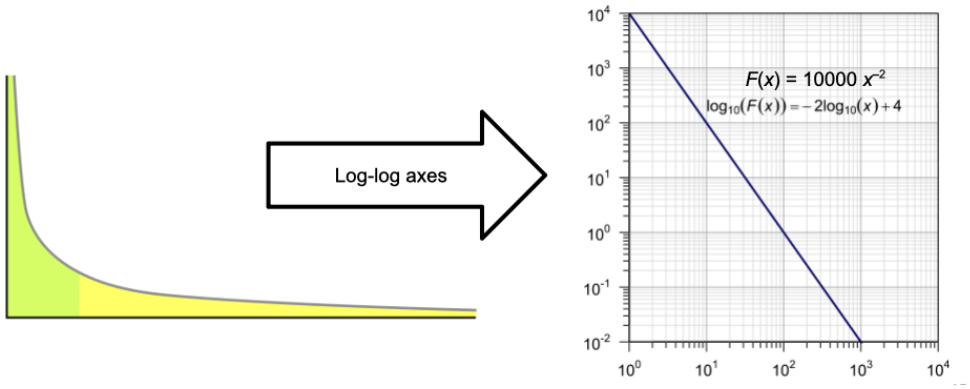
(Smoothed) histogram


Quantile-quantile (QQ) plots

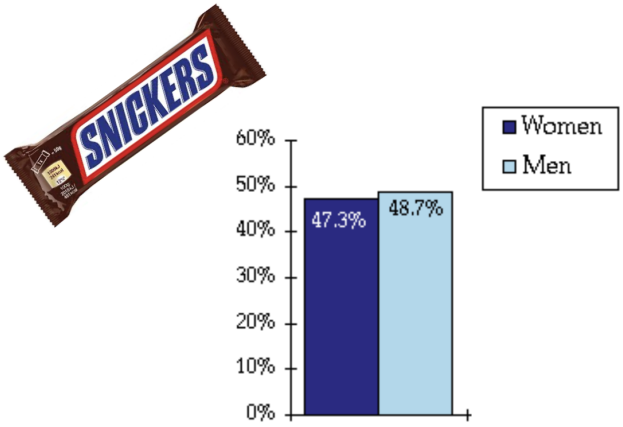
- Statistical tests:
  - Goodness-of-fit tests
  - Kolmogorov-Smirnoff test
  - Normality tests

Recognizing a power law

Use log-log axes, this is the best way to identify it. The result must be a decreasing linear line.

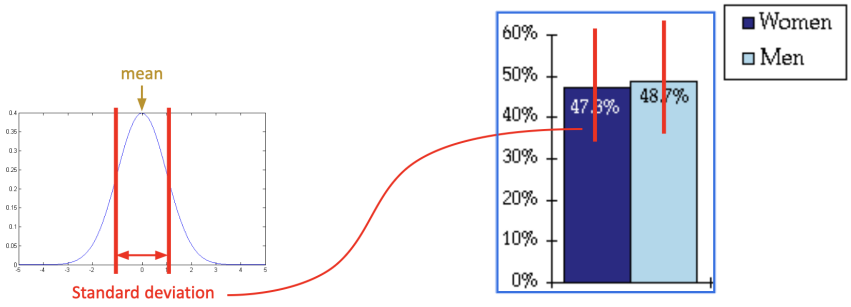


Study case: who likes snickers better?



Part 2: Quantifying uncertainty

- Most straightforward descriptive statistic to answer this question: mean for each group (women, men)



Always be curtains to quantify your uncertainty!

- Finite samples introduce uncertainty
  - Even a complete dataset is a finite sample!
- Whenever you report a statistic, you need to quantify how certain you are in it!
- We will discuss two ways of quantifying uncertainty:
  - (1) Hypothesis testing
  - (2) Confidence interval
- All plots should have error bars!

How to quantify uncertainty?

Approach 1: Hypothesis testing

Joseph B. Rhine ran an experiment where subjects had to guess whether 10 hidden cards were red or blue.

He found that about 1 person in 1000 had ESP ("extrasensory perception"), i.e, they could guess the color of all 10 cards!

Q: Do you agree? $0.5^{10} = 1/1024$ probability to guess correctly!

If there is no real effect, how likely is that I observer data as extreme as I observe?

$$1 - (1 - 2^{-10})^{1000} = 62\%.$$

This is the probability of at least someone guess the 10 cards correctly. So it's more likely that someone will guess than no one will guess it. He called back the "psychic" subjects and had them do the same test again. They all failed. He concluded that **the act of telling psychics that they have psychic abilities** causes them to lose...

Hypothesis testing

- A huge subject; can take entire classes on it
- Need to understand basics even if you don't use it yourself

- Never use it without understanding exactly what you're doing

The logic of hypothesis testing

- Flip a coin 100 times; outcome: 40 heads; "Is the coin fair?"
- Null hypothesis: "yes"; alternative hypothesis: "no"
- "How likely would I be to see an outcome at least this extreme (i.e., $\leq$ 40 heads) if the null hypothesis were true (i.e., if the coin were fair, i.e., if we expect 500 heads)?
- If this probability is large, the null hypothesis suffices to explain the data (and is thus not rejected)
- Otherwise, dig deeper in order to understand your data
- Idea: Gain (weak and indirect) support for a hypothesis H_A by **rulling out a null hypothesis** H_0
  - H_0: "the coin is fair"(simplest hypothesis)
  - H_A: "the coin is not fair (a.k.a. biased)"
- by inspecting a **test-statistic**: a measurement made on the data that is likely to be large under H_A but small under H_0
  - e.g., number of heads after 100 coin tosses (1-tailed)
  - e.g., abs(50 - number of heads after 100 coin tosses) (2-tailed)

Coin example (cont'd)

- **Null hypothesis** H_0: "the coin is fair", i.e., "probability of heads = 0.5"
- **Test statistic** s: abs(50 - of empirically observed heads after 100 coin tosses)
- Pr(S | H_0): probability distribution of test statistic, assuming that H_0 is true
- Decision rule: **reject** H_0 **if Pr(S** $\geq$ **s** | H_0**)** $<$ **a**,

  i.e., if the probability of deviating from 50 heads at least as much as empirically observed is small
  - Pr(S $\geq$ s | H_0) = **"p-value"**
  - a = **"significance level"**
- a controls "false-rejection rate" (probability of rejecting H_0 although it is true)
  - You as the data analyst choose a (common values: 5%, 1%, 0.5%, 0.1%)
  - Higher a $\Rightarrow$ higher false-rejection rate

Selecting the right test

There are many statistical tests. Although they differ in their details, the basic logic is always the same. The right choice of test depends on multiple factors (here a selection):

- Question

- Data type (continuous vs. categorical; dimensionality; number of outcomes)
- Sample size
- When comparing two samples: same population or different population?
- Parametric assumptions about distribution of test statistic under null hypothesis? (Less important for large sample size, due to central limit theorem)

Good news: **Plenty of advice available**

Remarks on p-values

- Widely used in all sciences
- They are widely misunderstood!
- Don't use them if you don't understand them!
- Large p means that even under a simple null hypothesis your data would be quite likely
- This tells you nothing about the alternative hypothesis
- Historically, not means as a method for formally deciding whether a hypothesis is true or not
- Rather an informal tool for assessing a particular result
- Low p-value means: "The simple null hypothesis doesn't explain the data, so keep looking for other explanations!
- p = 0.05 means: if you repeat the experiment 20 times, you'll see extreme data even under null hypothesis ⇒ you might have "lucked out"
- Look at the effect size ("y-axis"), not just the p-value!

Alternative approach: Bayes factors

$$\frac{Prob(Data, under H_0)}{Prob(Data, under H_A)}.$$

How to quantify uncertainty?

Approach 2: Confidence internals

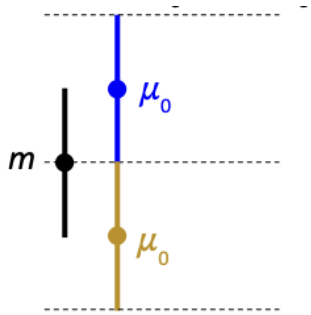
Confidence intervals: idea

- Confidence interval (CI) = a range of estimates for the parameter of interest (e.g., mean) that seems reasonable given the observed data
- Confidence level γ ⇒ "γ CI" (often $\gamma = 95\% \Rightarrow 95\% CI$)

Confidence intervals: definition

- μ: true value of parameter of interest
- m: empirical estimate of parameter of interest
- CIs and hypothesis testing are tightly connected:

– γ CI contains values μ_0 for which the null hypothesis "$H_0 : \mu = \mu_0$" cannot be rejected at significance level $1 - \gamma$

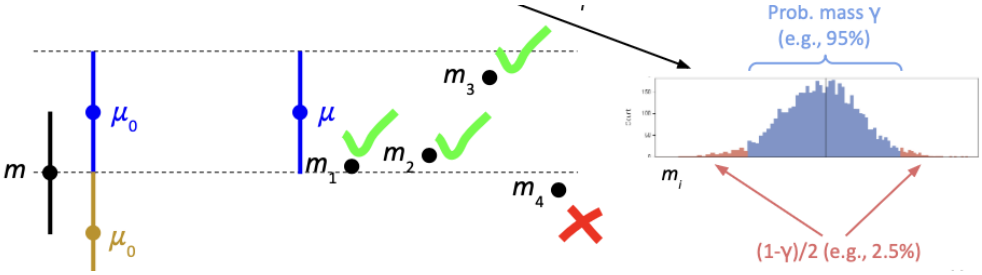


How to compute confidence intervals?

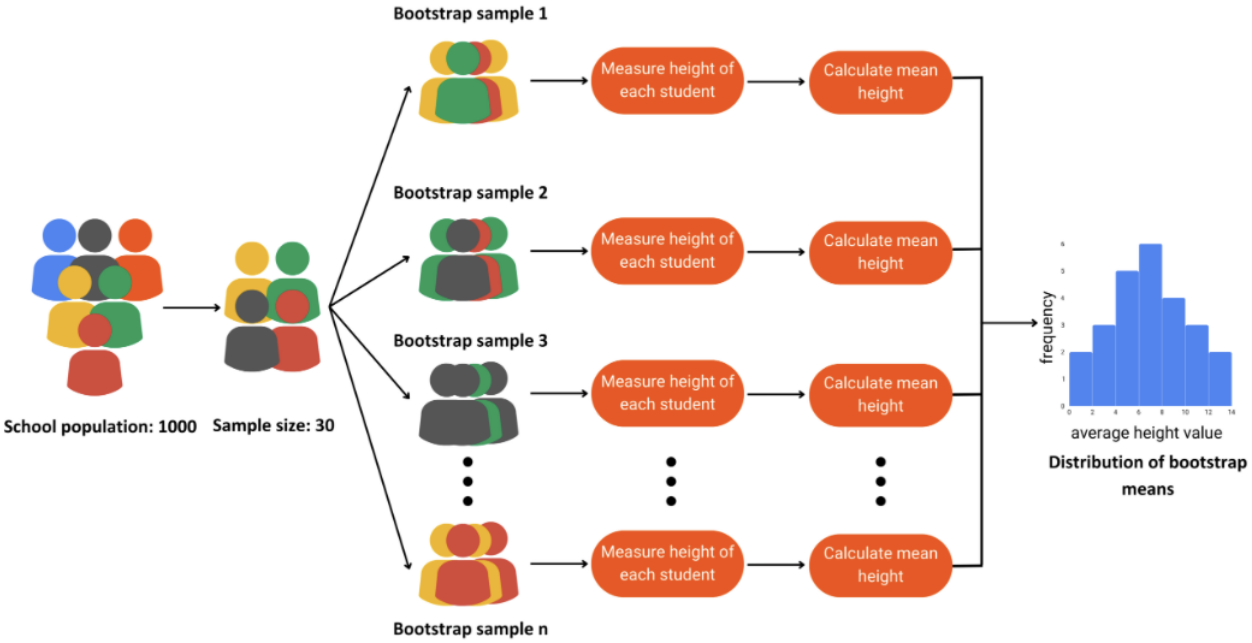
- **Parametric** methods assume that the test statistic follows a known (typically Normal) distribution ⇒ Need to verify that this is actually true! Ugh...
- **Non-parametric** methods make no assumptions about the distribution of the test statistic. They instead work by sampling the empirical data (Yay!)

Confidence intervals: another view

- If we were to repeat the data collection $N \Rightarrow \infty$ independent times, we'd obtains N estimates of $\mu : m_1, ..., m_N$
- Average of m_i's will approach the true μ (by the law of large numbers)
- For a fraction γ of the N repetitions, m_i lies within the γ CI around μ
- ⇒ May estimate CI from histogram of m_i's



Non-parametric CIs: bootstrap resampling



Bootstrap resamples: (random sampling with replacement) ⇒ Bootstrap statistics

Error bars

- An important use case for CIs
- But be careful! Error bars can potentially represent many things:
  - Confidence intervals (CIs)
  - Standard deviation
  - Standard error of the mean: standard deviation/sqrt(n)
- Always ask, always tell what the CIs represent!

Multiple-hypothesis testing

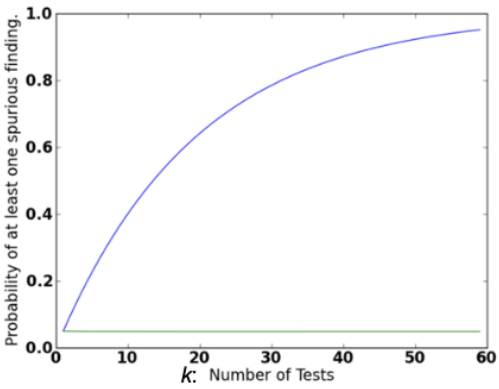
- If you perform experiments over and over, you're bound to find something
- If you consider "at least one positive outcome" to be the manifestation of an underlying effect: significance level must be adjusted down when performing multiple hypothesis tests!

P(detecting an effect when there is none) $= \alpha = 0.05$

P(detecting no effect when there is none) $= 1 - \alpha$

P(detecting no effect when there is none, on every experiment) $= (1 - \alpha)^k$

P(detecting an effect when there is none at least one experiment) $= 1 - (1 - \alpha)^k$



$\alpha = 0.05$

"Familywise Error Rate"

Bonferroni Correction

- Just divide by the number of hypothesis

$$\alpha_c = \frac{\alpha}{k}.$$

Sidak Correction

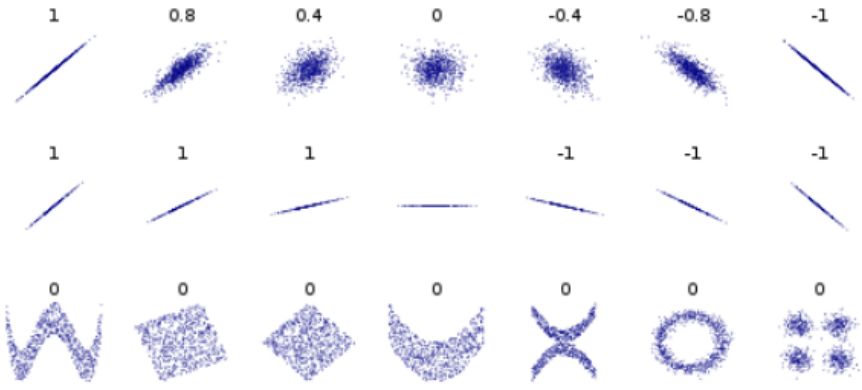
- Asserts independence

$$\alpha = 1 - (1 - \alpha_c)^k.$$
$$\alpha_c = 1 - (1 - \alpha)^{\frac{1}{k}}.$$

Part 3: Relating two variables

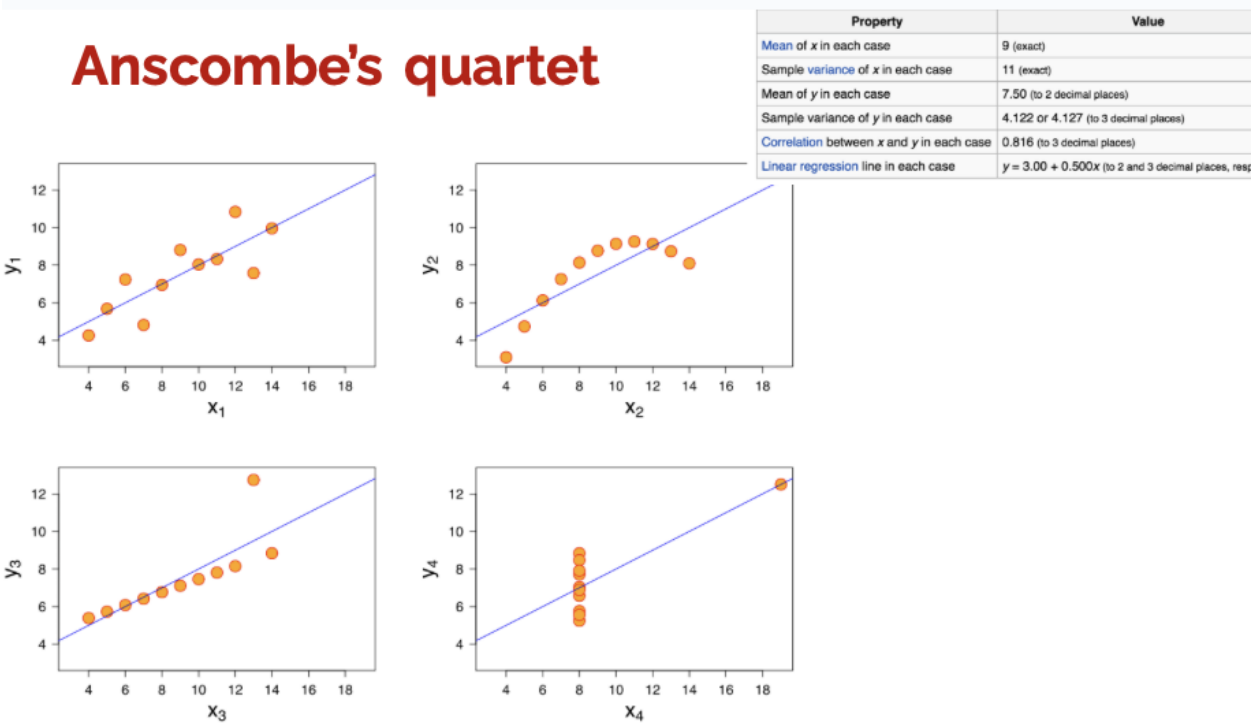
Pearson's correlation coefficient

- "Amount of linear dependence"



Note: some of these images are correlated but not linearly correlated...

- More general:
  - Rank correlation, e.g., Spearman's correlation coefficient
  - Mutual information
  - Correlation != causation

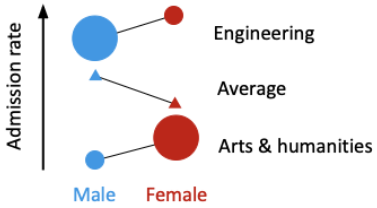


All different samples are equally correlated!

UC Berkeley gender bias (?)

Admission figures from 1973

	Applicants	Admitted
Men	8442	**44%**
Women	4321	35%



Department	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	825	62%	108	**82%**
B	560	63%	25	**68%**
C	325	**37%**	593	34%
D	417	33%	375	**35%**
E	191	**28%**	393	24%
F	373	6%	341	**7%**

Simpson's paradox

When a trend appears in different groups of data **but disappears or reverses when these groups are combined** – beware of aggregates!

In the previous example, **women tended to apply to competitive departments** with low rates of admission